RIWA KARAM

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Irvine, CA

Professional Summary

PhD candidate in Electrical and Computer Engineering specializing in multi-agent systems, robotics, machine learning, and optimization. Experienced in developing data-driven models, designing collaborative robotic systems, and translating research into practical solutions. Seeking research and engineering opportunities that advance intelligent autonomous systems through rigorous analysis, experimentation, and effective teamwork.

Industry Experience

- **ASML** [] — *June 2025 – September 2025*
  Mechatronics Research Engineer Intern — Wilton, CT
  - Investigated machine health monitoring using statistical and Bayesian hierarchical models for uncertainty quantification and structural analysis.
  - Developed a Bayesian hierarchical model using Python and PyMC to predict drift and performance degradation, enabling root-cause analysis across machine lifetimes.
  - Built and compared decision tree, random forest, multiple regression, and baseline regression models to evaluate performance and identify patterns in the data.
  - Conducted data collection, preprocessing, and analysis using Python and MATLAB, including regression modeling, principal component analysis (PCA), and heatmap-based pattern recognition.
  - Applied Markov chain Monte Carlo (MCMC), Bayesian inference, and time-lag regression methods to characterize model uncertainty and temporal behavior.
- **ASML** [] — *June 2024 – September 2024*
  Mechatronics and Control Systems Engineer Intern — Wilton, CT
  - Developed a Gaussian process regression (GPR) model using Python and MATLAB to predict force generated by a piezoelectric actuator with accuracy within ± 2 N; the work contributed to an ASML invention.
  - Built regression and machine-learning models to identify correlations, dependencies, and performance drivers in experimental data.
  - Automated and merged MATLAB and Python scripts that run testing workflows for both the mechatronics and software teams, reducing runtime from 40 hours to an overnight run.
  - Investigated feasibility of optimization problems and current-control hardware issues for test stands by analyzing their electrical circuitry and alternative hardware.
- **AviaPro Consulting Inc.** [] — *June 2022 – July 2023*
  Software Developer — Remote
  - Developed web applications using React.js and Node.js, replacing Excel-based repositories with streamlined interfaces connected to multiple databases.
  - Applied software-development and DevOps practices, including CI/CD, infrastructure as code, and Git.
  - Contributed to team-based full-stack projects and independently developed an internal full-stack application.
  - Transitioned to a part-time role in September 2022 after completing a summer co-op.

Academic Experience

- **UCI Robot Ecology Laboratory** [] — *September 2023 - Present*
  Graduate Student Researcher — Irvine, CA
  - Conduct research in collaborative multi-agent systems, optimization, machine learning, ecology-inspired robotics, coverage control, formation control, and human-swarm interaction under the supervision of Professor Magnus Egerstedt and Professor Yanning Shen.
  - Maintain the Robot Ecology Lab Robotarium testbed through robot testing, debugging, repair, and system troubleshooting.
  - Presented a first-author regular paper at the 64th IEEE Conference on Decision and Control (CDC) in Rio de Janeiro, Brazil.
  - Presented research findings at the 45th Southern California Control Workshop at the University of California, San Diego.
  - Passed my PhD qualifying examination in May 2026 and advanced to PhD candidacy.
  - Won the Software & Algorithms track at the 2026 UCI GEECS Annual Technology Showcase.
  - Presented a poster on my master's thesis at the 2025 UCI GEECS Annual Technology Showcase.
  - Mentor undergraduate researchers and collaborate with them on related research projects.
  - Co-authored the Robot Ecology Lab manual, documenting debugging procedures, experiment workflows, and Vicon system maintenance.

∘ Led laboratory tours and live multi-robot demonstrations for more than 100 visitors.

Education

- **University of California, Irvine** [] *September 2023 - Expected December 2027*
  Electrical & Computer Engineering, Doctor of Philosophy Irvine, CA
  - Dissertation: Collaboration in Multi-Agent Systems: From Assignment to Learning
  - Advisors: Professor Magnus Egerstedt, Professor Yanning Shen
- **University of California, Irvine** [] *September 2023 - June 2025*
  Electrical & Computer Engineering, Master of Science Irvine, CA
  - GPA: 3.975/4.00
  - Thesis: A Graphical Interface for Specifying and Establishing Multi-Robot Formations
  - Relevant Coursework: Optimization, Network Science, Control and Machine Learning, Linear Systems
- **University of Balamand, Al-Kurah** [] *August 2020 - June 2023*
  Computer Engineering, Bachelor of Science Balamand, Al-Kurah, Lebanon
  - GPA: 3.90/4.00
  - Computer Engineering Class of 2023 Valedictorian
  - Graduation Project: Gamification of Virtual Reality Kitchen Scenarios
  - Tutored peers through group and private sessions in courses such as programming (Python, MATLAB, C), Signals and Systems, Microcontrollers, Embedded Systems, and Computer Architecture.

Publications

C=Conference, T=Thesis, S=Submitted

[S.1] Riwa Karam, Ruoyu Lin, Brooks A. Butler, and Magnus Egerstedt, **Learning Altruistic Collaboration in Heterogeneous Multi-Team Systems** arXiv preprint, DOI.

[S.2] Riwa Karam, Alexander A. Nguyen, Ruoyu Lin, David R. Martin, Diana Morales, Brooks A. Butler, and Magnus Egerstedt, **Collaboration in Multi-Robot Systems: Taxonomy and Survey over Frameworks for Collaboration.** arXiv preprint, DOI.

[C.1] Riwa Karam, Ruoyu Lin, Brooks A. Butler, and Magnus Egerstedt, **Resource Allocation for Multi-Team Collaboration Based on Hamilton's Rule**, IEEE 64th Conference on Decision and Control (CDC), Rio de Janeiro, Brazil, 2025, pp. 6891-6898, DOI.

[T.1] Riwa Karam, **A Graphical Interface for Specifying and Establishing Multi-Robot Formations**, M.S. Thesis, University of California, Irvine, 2025, Official Record.

Skills

- **Programming Languages and Libraries:** MATLAB, Python, scikit-learn, PyTorch, Keras, NumPy, pandas, Matplotlib, shell scripting, PyMC, C, Java, C#.
- **Hardware Technologies:** Vicon Motion Capture (MoCap) System, robot testbed, robot building & debugging.
- **Other Tools & Technologies:** Git & GitHub, LaTeX, Simulink, Visual Studio Code, Unity, Microsoft Office, Google Workspace.

Selected Honors and Awards

- **GEECS Annual Technology Showcase Winner** *June 2026*
  Software & Algorithms Track, University of California, Irvine []
- **Faculty Award for Academic Excellence** *July 2023*
  University of Balamand, Faculty of Engineering []
- **First Place Winner (1st out of 12 teams), Lebanon IoT & AI Challenge 2021** *November 2021*
  Arab IoT & AI Challenge []
- **Dean's Honor List** *August 2020 - June 2023*
  University of Balamand, Faculty of Engineering

Professional Service

- **UCI Society of Graduate Electrical Engineers and Computer Scientists** [] *June 2026 – Present*
  Web Content Manager Irvine, CA
- **IEEE Control Systems Society NextCom** [] *March 2026 – Present*
  General Activities Committee Member Remote

Languages

- Arabic: Native proficiency
- English: Full professional proficiency
- French: Full professional proficiency
- Spanish: Elementary proficiency (CEFR A2)